

User-centric Reflective Writing Assistance: Leveraging RAG for Enhanced Personalized Support

Seyed Parsa Neshaei*
EPFL

Lausanne, Switzerland
seyed.neshaei@epfl.ch

Matea Tashkovska*
EPFL

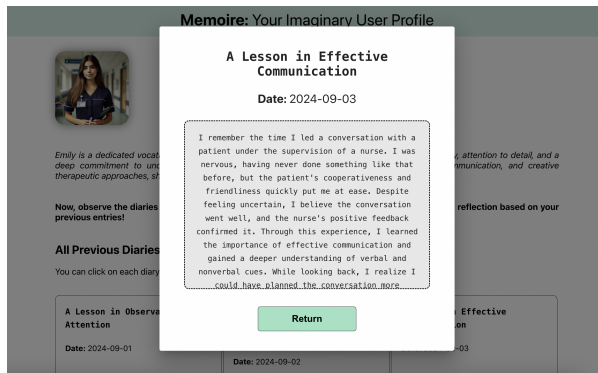
Lausanne, Switzerland
matea.tashkovska@epfl.ch

Paola Mejia-Domenzain
EPFL

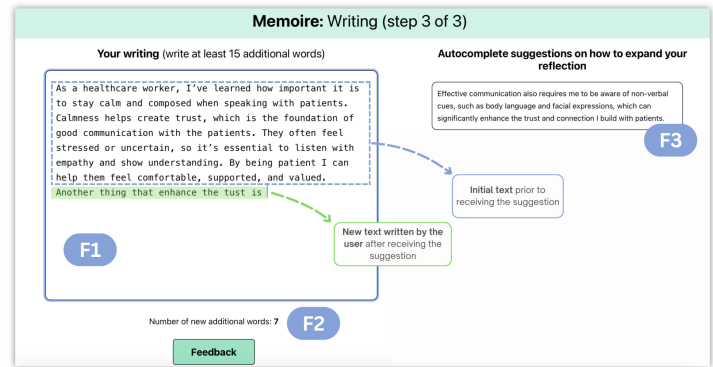
Lausanne, Switzerland
paola.mejia@epfl.ch

Thiemo Wambsganss
Institute for Digital Technology
Management, Bern University of
Applied Sciences
Bern, Switzerland
thiemo.wambsganss@bfh.ch

Tanja Käser
EPFL
Lausanne, Switzerland
tanja.kaeser@epfl.ch



(a)



(b)

Figure 1: An overview of the primary interface of Memoire, our intelligent assistant for reflective writing. Learners start by viewing their past written reflections (a) and then write new reflections with the support of intelligent, context-aware, and personalized suggestions generated using a RAG-based pipeline, related to both their most similar previous reflections and their current text (b). We have translated an imaginary user profile from German into English to present in this paper.

Abstract

Reflective writing, a valuable educational practice, helps learners analyze their thoughts and experiences. However, learners often struggle to translate their experiences and thoughts into written form. Recent advancements in large language models (LLMs) have demonstrated their potential for writing assistance and learning support, but they lack personalization, as their responses are not typically grounded in the personal experiences and prior knowledge states of individual learners. Retrieval-augmented generation (RAG) offers the potential to enable LLMs to incorporate learners' past data to provide personalized support. Despite potential, RAG has

yet to be fully evaluated for reflective writing. In this late-breaking work, we introduce Memoire, a reflective writing assistant using RAG to provide learners with personalized suggestions. We evaluate the effectiveness of our tool on 100 students in a classroom study, comparing three types of suggestions identified in prior research. Our study provides early insights into the potential of RAG-based methods to support reflective writing among students.¹

CCS Concepts

• Human-centered computing → Natural language interfaces; Field studies.

Keywords

reflective writing, intelligent writing assistants, retrieval-augmented generation, large language models

*Both authors contributed equally to this research.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3719899>

¹This work is an extension of our previous work-in-progress paper in the Second International Workshop on Generative AI for Learning Analytics (GenAI-LA) at LAK 2025.

ACM Reference Format:

Seyed Parsa Neshaei, Matea Tashkovska, Paola Mejia-Domenzain, Thiemo Wambsganss, and Tanja Käser. 2025. User-centric Reflective Writing Assistance: Leveraging RAG for Enhanced Personalized Support. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3706599.3719899>

1 Introduction and Related Work

Reflective writing is a metacognitive practice that encourages learners to analyze and document their thoughts, feelings, and insights about events or experiences. It is widely recognized in prior work as an effective approach to enhance self-regulation, deepen understanding, and foster critical thinking [18, 35]. Reflective writing has demonstrated significant benefits across educational and professional contexts, including improving learning outcomes [18], supporting professional development [6], and helping individuals adapt their future behavior in similar situations [24]. In some educational settings, such as vocational education, which combines theoretical learning with practical workplace experiences, reflective writing is particularly important for bridging the gap between classroom concepts and hands-on application [5, 25].

Despite its proven benefits, reflective writing presents challenges for many learners, particularly those new to this practice. Research shows that inexperienced writers often struggle to produce coherent and meaningful reflections and find it difficult to synthesize their emotions, beliefs, and experiences into structured texts [1, 17]. Moreover, a common challenge faced by student writers, even those with satisfactory grammar and spelling skills, is writer's block, which limits their ability to convey ideas coherently [20]. To address the challenges of effective writing support, human-computer interaction (HCI) researchers have proposed intelligent writing assistants [13]. While these tools hold promise, most existing solutions focus on providing generic feedback and fail to offer personalized support tailored to individual learners' prior experiences and knowledge, especially in the domains of metacognition and reflective writing. Personalization is crucial, as students learn more effectively when new concepts are contextualized within their own experiences and knowledge base. Although prior studies have explored writing assistants for reflective writing [12, 37], there is a notable gap in leveraging learners' prior reflective entries and personal knowledge [22] to deliver tailored, intelligent support. The emergence of large language models (LLMs) has brought significant advancements in writing assistance, offering intelligent feedback across diverse domains such as academic writing [9], story writing [38], and text summarization [7]. However, despite their potential, LLM-based tools also face limitations, particularly a lack of personalization [3]. Most LLM-driven approaches focus on analyzing and providing feedback solely on the current piece of writing [10, 33, 34], neglecting the opportunity to incorporate learners' prior entries as a foundation for generating more meaningful and personalized suggestions.

A promising approach to mitigate these issues, which we employ in our study, is Retrieval-Augmented Generation (RAG), a method that enhances LLM outputs by incorporating external data in the generation process [14]. Incorporating RAG into a writing assistant enhances the accuracy of outputs by using user-specific data

retrieved from relevant data sources, thereby providing suggestions aligned with each learner's unique experiences. RAG-based approaches not only enable LLMs with smaller context windows to incorporate external information, but also mitigate the *diminished focus* on relevant information, when used with LLMs bearing larger context windows [39]. Although prior works in education, such as comprehension of scientific texts [8, 30], have utilized RAG, its application in reflective writing and metacognition tools to enhance personalized support remains underexplored. Moreover, there is a lack of work on exploring which support modalities and suggestion types are most effective for writing assistants to assist with reflective writing. Specifically, questions remain about how different types of suggestions impact learners' ability to reflect deeply and improve their writing quality. This highlights the need for research into identifying the most impactful approaches to support learners in reflective writing. To address these research gaps, we propose integrating a RAG-based pipeline into a writing assistant for reflective writing. Specifically, we aim to answer the following research question: **(RQ)** How do learners perceive the usefulness and effectiveness of a reflective writing assistant that provides personalized RAG-based suggestions?

To answer our research question, we introduce Memoire, our intelligent RAG-based reflective writing assistant. Memoire leverages a RAG-based approach to deliver adaptive and personalized suggestions for improving and expanding the learner's current reflection, based on their prior reflections. Specifically, our system incorporates three types of support inspired by prior research: critical questions [16, 26], autocomplete suggestions [13, 28], and summarizing feedback [7, 36], with the aim of identifying the most effective modality to assist learners in reflective writing. To evaluate Memoire, we conducted a classroom study with 100 students from a Western European vocational school, where we collected both quantitative and qualitative feedback on comparing suggestion type, as well as their overall perceptions of the system. Our initial evaluation revealed a high perceived ease of use and usefulness among students, with participants recognizing Memoire's potential to support their reflective writing. Notably, students found the critical question suggestions most helpful, as they encouraged deeper thinking and facilitated meaningful reflection. In contrast, summarizing feedback was often considered less relevant and less effective in improving their writing. Participants also valued the system's ability to connect their prior experiences to current reflections, which they found particularly meaningful. These results highlight the potential of RAG-based methods to make reflective writing tools more context-aware and personalized, and further support their integration into educational settings.

Our work introduces several key contributions to the research direction of integrating intelligent writing assistants into reflective learning processes. Firstly, we introduced a novel application of RAG for reflective writing tasks, addressing the challenge of providing personalized and context-aware suggestions by leveraging learners' prior written reflections. Secondly, we implemented and compared three distinct RAG-based suggestion types to identify the most effective approaches for enhancing reflective writing and user engagement. Lastly, we integrated our RAG pipeline and suggestion types into Memoire, our RAG-based reflective writing assistant, and evaluated its impact through a single-intervention classroom study

with vocational students, providing early insights into its usability and the perceived effectiveness of the different suggestion types in a real-world educational setting. Our work paves the way for a larger-scale study of Memoire among vocational school students, e.g., a longitudinal study across a semester, to compare the benefits of using Memoire on learning gains and reflective writing quality versus not receiving any RAG-based AI support.

2 Memoire: Implementing RAG into Reflective Writing Assistance

Our system, Memoire, is designed to assist users in reflective writing by providing suggestions based on their previous reflections. In the following section, we describe the design, functionalities, and implementation of Memoire.

2.1 Interaction Design of Memoire

The main interface of Memoire, developed using the React framework and inspired by previous works in writing assistance [13], can be seen in Figure 1. Users first see their latest 15 written reflections, which they can open and re-read (Figure 1-a). These reflections also form each user's data source for the RAG component of Memoire (described in Section 2.3). Then, users enter the *writing* page (Figure 1-b), where they can write their new reflection in the text field on the left side (F1). We added a caption (F2) below the text area that shows the live number of words typed in the text area, following a range of prior writing assistants [19, 34] and allowing us to impose a minimum word count in the experimental study. After writing their reflection, users receive suggestions (F3) from one of the three suggestion types (described in Section 2.2). These suggestions are relevant to the context of their currently written text and their most similar prior reflection provided by the RAG pipeline. We followed the standards established in prior writing assistants [13] to implement the suggestions in our design. Learners can continue editing their text based on the received suggestions and can also get additional suggestions on their updated text until they are satisfied with their writing.

2.2 Suggestion Types

To determine the types of suggestions to present to users in Memoire, we explored the literature on HCI, learning sciences, and writing assistants. Our objective was to identify suggestion types that have been effectively utilized and validated in previous works to enhance the writing process. By integrating these suggestion types into Memoire, we aim to provide users with contextually relevant and meaningful feedback that can improve the quality of their reflections. Moreover, these suggestions are specifically designed to address common challenges faced during reflective writing, including generating ideas, overcoming writer's block, improving clarity, and fostering deeper engagement with their thoughts. More precisely, we selected three distinct types of suggestions that are particularly well-suited to support reflective writing:

1) Critical Questions: Previous works have considered asking questions as a method to stimulate critical thinking and have shown that this approach improves students' learning outcomes [11, 16, 26]. Inspired by these findings, users were presented with three questions designed to help them maintain a thoughtful and

reflective tone in their writing. The questions were presented in a numbered list and were tailored to align with the content that the users had already written. The goal of the critical questions is to inspire users to engage more deeply with their reflections and explore their thoughts more comprehensively, rather than just provide direct answers. This approach aligns with the principles of reflective writing, which emphasize examining thoughts, emotions, and actions to gain deeper insights [35].

2) Autocomplete Suggestions: Giving ideas to help writers complete their text has been explored in the literature before [13, 23, 28]. The autocomplete suggestion provides users with a single sentence to assist them in continuing their writing. The sentence is generated based on the context of the text users have already written and their most similar previous reflection, to ensure that the suggestion is coherent and meaningful. The goal of this suggestion is to reduce the cognitive load on users by offering a sentence they can build upon, helping them overcome potential writing blocks. This type of assistance is particularly beneficial in situations where users may struggle to formulate their next thoughts or organize them cohesively.

3) Summary Feedback: Finally, Memoire includes a module for summarizing feedback, as explored in prior research [7, 36]. This type of suggestion provides feedback on the current text by summarizing content from prior reflections that are contextually aligned with the current writing. The generated summary focuses on key ideas, insights, and lessons from the most relevant past reflections, enabling learners to draw inspiration from their earlier work and integrate past experiences into their current reflection. This process aims to help learners connect new insights with their prior knowledge.

2.3 RAG Implementation

Memoire leverages RAG to provide contextually relevant suggestions to the users. RAG enhances language model outputs by referencing an external knowledge base, which in this case consists of the user's prior reflections, to ensure that the suggestions are accurate, reliable, and grounded in their previous reflections. The RAG system consists of two primary components:

Retriever. This component in our RAG system was designed to efficiently select past reflections that are most relevant to the student's current writing. To achieve this, we used OpenAI's *text-embedding-3-small* model, chosen for its optimization for low latency and its ability to handle inputs up to 8,191 tokens. This enabled us to process longer reflections while ensuring the system remains responsive. We considered every prior reflection of a user as a document available for retrieval. Each past reflection was converted into a 1536-dimensional embedding vector, creating a vector database of semantically meaningful representations. This database serves as the foundation for our retrieval system. Every time the student is writing a reflection, the current text is embedded using the same model and treated as the query. To measure the relevance between the query and each past reflection, we used cosine similarity.

Generator. For the generator component, we experimented with the latest versions of both LLaMA [31] and GPT [21] models. While LLaMA offered the advantage of being open-source, we ultimately

chose the GPT-4o model provided by OpenAI due to its faster response time with respect to our resources, as well as the higher accuracy of its generated outputs in our task, as observed during development. This model is consistently used by researchers for the quality of its outputs and is also adopted in existing intelligent assistants [29]. Each time the student writes a reflection, the generator uses the embedded query and the retrieved past reflections, along with the prompt, to produce a concise and contextually relevant response. To refine and evaluate the quality of the prompts, we conducted a human evaluation process involving three researchers in HCI and AI for education. These researchers assessed the relevance, coherence, and overall quality of the outputs. This process allowed us to ensure the system consistently delivers high-quality suggestions that are tailored to the users' needs. The best-performing prompts for each suggestion type can be found in the Appendix.

3 Classroom Experimental Evaluation

To gain early insights on how learners perceive a reflective writing assistant with RAG-based suggestions personalized to their prior experiences, we conducted a classroom evaluation of Memoire in six nursing and caring apprenticeship classrooms among 100 students (88 female, 12 male², average age = 23.28, $SD = 4.80$) at a German-speaking vocational school in a Western European country. All students provided informed consent to participate in our research. They all regularly participated in reflective writing throughout the semester using a data recording web application without AI support, with an average of 26.96 reflections ($SD = 13.30$) per student. The prior reflections of students written during the semester were used as the data source for our RAG pipeline.

3.1 Study Design

Our study consisted of a pre-survey, the writing task, and a post-survey. Students were instructed to spend a maximum of 60 minutes on the study. An overview of the study design can be seen in Figure 2.

Pre-survey. In the pre-survey, we first collected the gender and age of the users. Then, we asked questions on two behavioral constructs from prior works [2, 4, 19, 32] measured in 1-7 Likert scale to ensure proper randomization: 1) *information technology usage model* (e.g., “If I heard about a new information technology, I would look for ways to experiment with it”), and 2) *feedback-seeking behavior* (e.g., “It is important for me to receive feedback on my performance”). These constructs were originally explored in prior research [2, 4] and have been applied in writing assistant contexts in recent studies [19, 32]. We also asked them about their previous participation in reflective writing. Our study did not find any significant difference in any pre-survey construct among users receiving different orders of writing tasks, ensuring correct randomization.

Writing Task. After the pre-survey, users entered the writing interface to write a new reflection. The interface displayed suggestions from each of the three suggested types (see Section 2.2) across three consecutive pages, in a randomized order for each user. Before receiving any suggestions, users wrote an initial reflection of at least 50 words, which was saved as a starting point

for the subsequent pages. After receiving each suggestion, users adapted their initial text by adding at least 25 words and had the chance to request more suggestions if needed. Finally, after each suggestion type, users were asked four 1-5 Likert-scale questions to enable comparison of different suggestion types later on: “I like the suggestions above,” “The suggestions above are relevant to my text,” “The suggestions above are helpful,” and “The suggestions above are accurate.”

Post-survey: In the final phase, students *ranked* the three suggestion types from most to least preferred using a drag-and-drop interface and provided written explanations of their reasoning. They also responded to a set of perception constructs, used in writing assistance research to assess the system's effectiveness and usability [19, 32], namely *technology acceptance*, *perceived usefulness*, *perceived ease of use*, *perceived improvement in writing*, *perceived improvement in writing in the long run*, *correctness of the suggestions*, and *excitement after interaction*. All constructs were measured on a 1- to 7-point Likert scale. Finally, participants were also asked to share open-ended feedback, enabling us to gain deeper insights into their experiences with Memoire.

4 Results

Perception towards Memoire. We found a general positive sentiment toward Memoire based on the post-survey metrics (signified by means higher than the neutral value of 4 in the 1-7 Likert scale). The distribution of responses for each construct is shown in Figure 3, illustrating that most participants selected options on the positive end of the Likert scale. The observed mean and SD per construct can be seen in the Appendix.

We also analyzed open-ended responses to gain deeper insights into student perception. Many shared a positive experience with Memoire, with 21 participants praising its intuitive interface and user-friendly design. Students described it as “simple and intuitive” and “well-structured,” showing its ease of use. Also, 19 participants recognized its potential to improve reflection, noting that it encouraged further thinking and linked theories to situations. Participants also valued its support for reflective writing, mentioning its ability to help “write detailed reflections” and “enhance reflective and precise writing.” Twelve respondents noted the potential of Memoire to provide “long-term benefits for reflective writing.” While most participants found Memoire effective, they highlighted areas for improvement, such as repetitive feedback and occasional irrelevance in suggestions. Despite these challenges, the overall feedback suggests the promise of Memoire as a valuable tool for reflective writing. Many students expressed interest in exploring its features and future improvements, including one student who requested to have the tool available for further use throughout the semester.

Finding the effective suggestion types. We analyzed participants' scores for each suggestion type, in terms of *liking*, *relevance*, *helpfulness*, and *accuracy*. A repeated-measures ANOVA with Benjamini-Hochberg correction revealed no statistically significant differences in any metric ($p > .05$). Thus, we found no clear evidence that participants consistently rated one suggestion type as more preferred, across the evaluated criteria. Therefore, we conducted a deeper analysis of students' comparative ranking of suggestion types in the post-survey to gain insights into their preferences. Our

²This was consistent with the general gender distribution of the students in the vocational school.

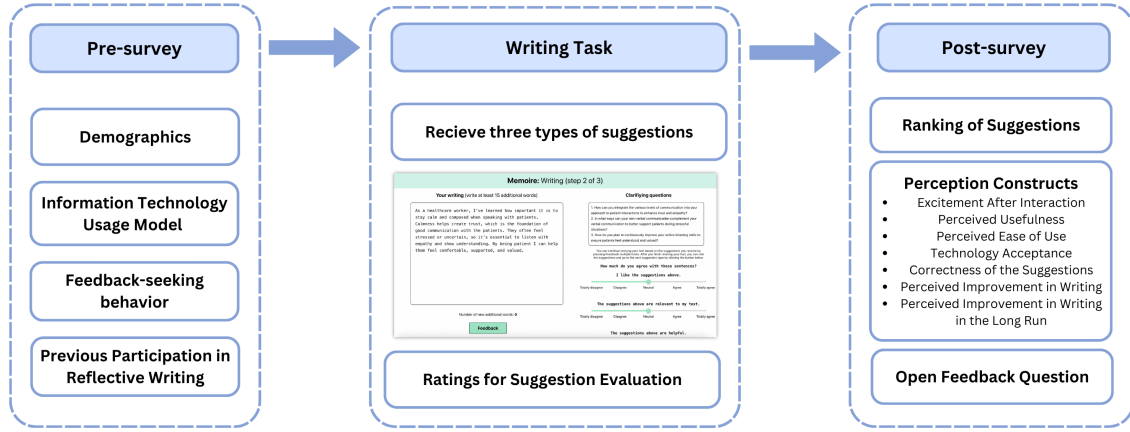


Figure 2: An overview of the study design of Memoire, consisting of a pre-survey to ensure randomization, a writing task encompassing the three suggestion types, and a post-survey to collect perception metrics.

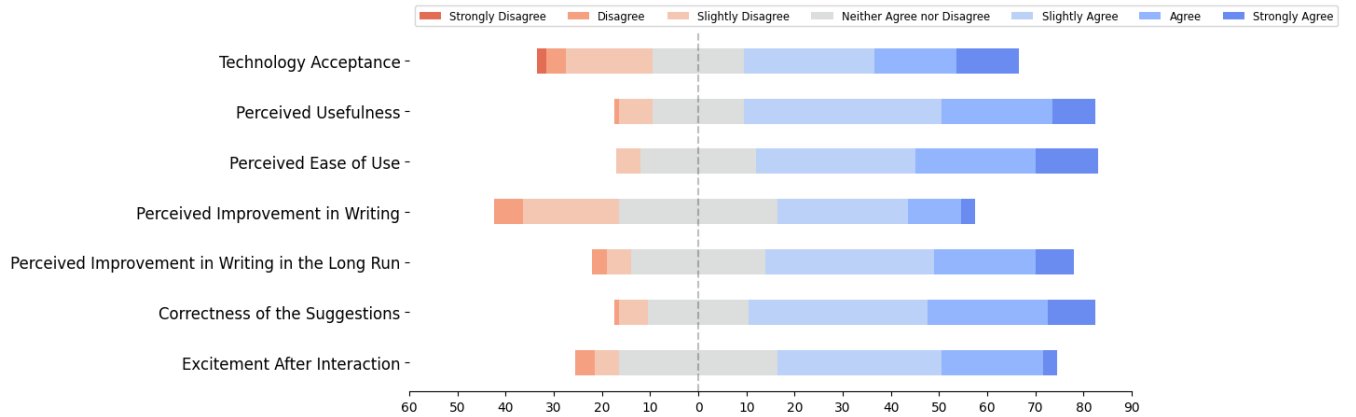


Figure 3: Distribution of participant responses across perception constructs measured on a 7-point Likert scale.

analysis revealed statistically significant differences, as indicated by the Friedman test ($\chi^2(2) = 9.62, p = .008$). Critical questions were the most preferred suggestion type, ranked first by 44 participants. Autocomplete ranked second with 31 participants, while summary was the least favored, with only 25 participants ranking them first and 49 ranking them third. Pairwise comparisons using the Wilcoxon test (with p-values corrected using the Benjamini-Hochberg procedure) showed significant differences between critical questions and summary ($p = .021$) and a marginally significant difference between autocomplete and summary ($p = .050$), with summary being less preferred in both cases. This highlights a preference for critical questions over other types, especially compared to summary suggestions.

To justify this finding, we explored participants' answers to the open-ended question about their rankings. The **critical questions** suggestions were found helpful for encouraging thought and reflection. 33 students mentioned they "helped them engage in deeper reflection" or "stimulated deeper thinking". The questions were also found to enhance writing flow by suggesting areas for exploration. This aligns with the goals of reflective writing. However,

4 participants found that the questions led to repetitive writing. **Autocomplete suggestions** was found to provide new ideas when students were unsure of what to write next. Students believed that these suggestions "filled gaps" in reflections, offered meaningful expansions, and "revealed additional aspects." However, 11 participants found them less helpful or contextually relevant, limiting their impact. **Summaries** were the least favored. 19 participants found them unhelpful due to redundancy, irrelevance, or inaccuracies. When summaries didn't align with the text, they felt confusing or misleading. Despite these criticisms, students acknowledged summaries could be beneficial for "reviewing ideas," recalling past writing, and making connections to past experiences.

5 Discussion and Conclusion

By integrating RAG into reflective writing, Memoire, our reflective writing assistant, addresses common challenges including writer's block and connecting new ideas to past experiences. Our classroom study showed positive perceptions of Memoire as a valuable

educational tool. Students found it easy to use and useful, supporting research showing that perceived ease of use and usefulness influence AI application adoption [15]. Students appreciated the system's ability to connect current reflections with past entries, enabling them to build on past experiences and knowledge. They also acknowledged Memoire's potential to enhance their reflective writing skills. They noticed immediate improvement and expected even greater long-term improvement with continued use, based on the responses to the post-survey constructs.

We also found that *critical questions* were the most preferred suggestion type among participants, as they encouraged deeper thinking and offered new perspectives for reflection. This aligns with reflective writing goals, which emphasize critically examining experiences for insights and connecting them to broader learning contexts. In contrast, *summary* suggestions were often seen as less helpful, possibly due to the limited relevance of summarized prior reflections. We believe this happened because some similar past reflections might not align closely enough with the current writing to provide useful suggestions. This limitation underscores the need for a larger dataset of prior reflections to fully realize the potential of RAG-based systems. Future implementations of Memoire could benefit from incorporating more frequent and longer-term reflections to improve suggestion quality and relevance.

Overall, our study provides initial evidence supporting the use of RAG and demonstrating its ability to address common issues in reflective writing. By connecting current reflections with past experiences and offering personalized support, Memoire fosters deeper, more meaningful engagement in the reflective writing process.

Limitations and Future Work. Nevertheless, our study comes with several limitations. First, the short duration and single-use interaction restrict assessing the long-term impact of Memoire on students' reflective writing, metacognitive development, and learning outcomes. We plan to conduct a longer study over a semester or academic year in future work, with more frequent use of Memoire to derive more robust insights. Moreover, we believe that reflective writing practices vary based on subject matter, culture, and educational environment. In specialized or underexplored domains, the LLM might struggle to provide meaningful or accurate suggestions.

Building on this study's findings, we propose several future research directions. Analyzing students' text at each step and examining how the learners integrated the suggestions into their texts could reveal insights into the tool's effectiveness. Also, motivated by participants' feedback that summary suggestions were less helpful due to insufficient past *relevant* reflections, future work could explore if the preferred suggestion type correlates with the similarity between the current reflection and the most similar past reflection retrieved by the RAG pipeline. This analysis could refine suggestion mechanisms to better align with user preferences and enhance the overall personalization of our tool. Additionally, future work can explore how writing assistants similar to Memoire affect the structure and writing quality of the reflections, as opposed to merely measuring user preference metrics. A more in-depth analysis of how each suggestion type is applied to the texts can also uncover insights on the balance between streamlining the writing process and deep engagement with prior experiences. Also, we call for future researchers to explore if using prior reflections to

provide suggestions can have any kind of an *echo chamber* effect on students, limiting novel or meaningful reflective writings.

We also note the differences between supporting the reflective *process* of students (i.e., the cognitive process within learners), versus merely generating better reflective *writings* (i.e., the external output). While using LLMs can reduce the mental efforts of learners, they have been shown to compromise the depth of conceptual engagement with the content [27], possibly limiting their applicability in reflection. We thus call for future researchers to explore the risks of employing LLMs in the context of reflective writing. Finally, we also call for future work to extend upon the technical implementation of Memoire, particularly comparing the RAG-based pipeline with a baseline of LLMs without RAG, to provide insights into how RAG-based approaches can provide additional value for personalization in reflective writing settings.

Conclusion. In this work, we introduced Memoire, an intelligent reflective writing assistant that provides personalized RAG-based suggestions to enhance learners' reflective writing. Memoire incorporates critical questions, autocomplete suggestions, and summarizing feedback as three suggestion types. A classroom study with 100 vocational school students showed that Memoire effectively promotes meaningful support in reflective writing, with participants valuing critical questions as most effective for fostering deeper reflection. Our research contributes insights into designing and implementing RAG-based systems in education, emphasizing personalization and context-aware suggestions. As generative AI advances, Memoire serves as a promising step in exploring how adaptive AI-based tools can support learners in reflective writing and promote critical thinking and metacognitive skills.

Acknowledgments

This project was substantially financed by the Swiss State Secretariat for Education, Research, and Innovation (SERI).

References

- [1] Ikin Syamsudin Adeani, R Bunga Febriani, and Syafryadin Syafryadin. 2020. Using GIBBS' reflective cycle in making reflections of literary analysis. *Indonesian EFL Journal* 6, 2 (2020), 139–148.
- [2] Ritu Agarwal and Elena Karahanna. 2000. Time flies when you're having fun: Cognitive absorption and beliefs about information technology usage. *MIS quarterly* (2000), 665–694. Publisher: JSTOR.
- [3] Ashraf Alam. 2023. The secret sauce of student success: Cracking the code by navigating the path to personalized learning with educational data mining. In *2023 2nd international conference on smart technologies and systems for next generation computing (ICSTSN)*. IEEE, 1–8.
- [4] Susan J Ashford. 1986. Feedback-seeking in individual adaptation: A resource perspective. *Academy of Management journal* 29, 3 (1986), 465–487. Publisher: Academy of Management Briarcliff Manor, NY 10510.
- [5] Alberto AP Cattaneo and Elisa Motta. 2021. "I reflect, therefore I am... a good professional". On the relationship between reflection-on-action, reflection-in-action and professional performance in vocational education. *Vocations and Learning* 14, 2 (2021), 185–204. Publisher: Springer.
- [6] Marilyn Cochran-Smith. 2005. Studying teacher education: What we know and need to know. Issue: 4 Pages: 301–306 Volume: 56.
- [7] Hai Dang, Karim Benharrah, Florian Lehmann, and Daniel Buschek. 2022. Beyond text generation: Supporting writers with continuous automatic text summaries. In *Proceedings of the 35th annual ACM symposium on user interface software and technology*. 1–13.
- [8] Michael Galarnyk, Rutwik Routu, Kosha Bheda, Priyanshu Mehta, Agam Shah, and Sudheer Chava. 2024. ACL ready: RAG based assistant for the ACL checklist. *arXiv preprint arXiv:2408.04675* (2024).
- [9] Katy Ilonka Gero, Vivian Liu, and Lydia Chilton. 2022. Sparks: Inspiration for science writing using language models. In *Proceedings of the 2022 ACM designing interactive systems conference*. 1002–1019.

- [10] Andreas Göldi, Thiemo Wambsganss, Seyed Parsa Neshaei, and Roman Rietsche. 2024. Intelligent support engages writers through relevant cognitive processes. In *Proceedings of the CHI conference on human factors in computing systems*. 1–12.
- [11] Yueh-Ren Ho, Bao-Yu Chen, and Chien-Ming Li. 2023. Thinking more wisely: using the Socratic method to develop critical thinking skills amongst healthcare students. *BMC medical education* 23, 1 (2023), 173. Publisher: Springer.
- [12] Taewan Kim, Seolyeong Bae, Hyun Ah Kim, Su-woo Lee, Hwajung Hong, Chanmo Yang, and Young-Ho Kim. 2024. MindfulDiary: Harnessing large language model to support psychiatric patients' journaling. In *Proceedings of the CHI conference on human factors in computing systems*. 1–20.
- [13] Mina Lee, Katy Ilonka Gero, John Joon Young Chung, Simon Buckingham Shum, Vipul Raheja, Hua Shen, Subhashini Venugopalan, Thiemo Wambsganss, David Zhou, Emad A Alghamdi, and others. 2024. A design space for intelligent and interactive writing assistants. In *Proceedings of the CHI conference on human factors in computing systems*. 1–35.
- [14] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, and others. 2020. Retrieval-augmented generation for knowledge-intensive NLP tasks. *Advances in Neural Information Processing Systems* 33 (2020), 9459–9474.
- [15] Reena Malik, Ambuj Shrama, Sonal Trivedi, and Rikkee Mishra. 2021. Adoption of chatbots for learning among university students: Role of perceived convenience and enhanced performance. *International Journal of Emerging Technologies in Learning (iJET)* 16, 18 (2021), 200–212. Publisher: International Journal of Emerging Technology in Learning.
- [16] Katia Martha. 2023. Improving critical thinking in written assignments: Human vs. ChatGPT tutor in socratic questioning intervention. (2023).
- [17] Rebekkah Middleton. 2017. Critical reflection: the struggle of a practice developer. (2017).
- [18] Jennifer A Moon. 2013. Reflection in learning and professional development: Theory and practice. (2013). Publisher: Routledge.
- [19] Seyed Parsa Neshaei, Roman Rietsche, Xiaotian Su, and Thiemo Wambsganss. 2024. Enhancing peer review with AI-powered suggestion generation assistance: Investigating the design dynamics. In *Proceedings of the 29th international conference on intelligent user interfaces*. 88–102.
- [20] Lawrence J Oliver. 1982. Helping students overcome writer's block. *Journal of Reading* 26, 2 (1982), 162–168. Publisher: JSTOR.
- [21] OpenAI. 2023. GPT-4 technical report. *arXiv preprint arXiv:2303.08774* (2023). <https://arxiv.org/abs/2303.08774>
- [22] Viktoria Pammer-Schindler and Michael Prilla. 2021. The reflection object: An activity-theory informed concept for designing for reflection. *Interacting with Computers* 33, 3 (2021), 295–310. Publisher: OUP.
- [23] Sourav Prosad, Viswa Datha Polavarapu, and Shrutendra Harsola. 2023. An ensemble approach to personalized real time predictive writing for experts. *arXiv preprint arXiv:2308.13576* (2023).
- [24] Mary Ryan and Michael Ryan. 2013. Theorising a model for teaching and assessing reflective learning in higher education. *Higher Education Research & Development* 32, 2 (2013), 244–257. Publisher: Taylor & Francis.
- [25] Beat A Schwendimann, Alberto AP Cattaneo, Jessica Dehler Zufferey, Jean-Luc Gurtner, Mireille Bétrancourt, and Pierre Dillenbourg. 2015. The 'Erfahrraum': A pedagogical model for designing educational technologies in dual vocational systems. *Journal of Vocational Education & Training* 67, 3 (2015), 367–396. Publisher: Taylor & Francis.
- [26] Kumar Shridhar, Jakub Macina, Mennatallah El-Assady, Tanmay Sinha, Manu Kapur, and Mrinmaya Sachan. 2022. Automatic generation of socratic subquestions for teaching math word problems. *arXiv preprint arXiv:2211.12835* (2022).
- [27] Matthias Stadler, Maria Bannert, and Michael Sailer. 2024. Cognitive ease at a cost: LLMs reduce mental effort but compromise depth in student scientific inquiry. *Computers in Human Behavior* 160 (2024), 108386. Publisher: Elsevier.
- [28] Xiaotian Su, Thiemo Wambsganss, Roman Rietsche, Seyed Parsa Neshaei, and Tanja Käser. 2023. Reviewer: AI-Generated instructions for peer review writing. In *Proceedings of the 18th workshop on innovative use of NLP for building educational applications (BEA 2023)*. 57–71.
- [29] Surendrabikram Thapa and Surabhi Adhikari. 2024. GPT-4o and multimodal large language models as companions for mental wellbeing. *Asian Journal of Psychiatry* 99 (2024), 104157. Publisher: Elsevier.
- [30] Dominik Thüs, Sarah Malone, and Roland Brünken. 2024. Exploring generative AI in higher education: a RAG system to enhance student engagement with scientific literature. *Frontiers in Psychology* 15 (2024), 1474892. Publisher: Frontiers Media SA.
- [31] Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, and others. 2023. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971* (2023).
- [32] Thiemo Wambsganss, Andreas Janson, and Jan Marco Leimeister. 2022. Enhancing argumentative writing with automated feedback and social comparison nudging. *Computers & Education* 191 (2022), 104644. Publisher: Elsevier.
- [33] Florian Weber, Thiemo Wambsganss, Seyed Parsa Neshaei, and Matthias Soellner. 2023. Structured persuasive writing support in legal education: A model and tool for German legal case solutions. In *Findings of the association for computational linguistics: ACL 2023*. 2296–2313.
- [34] Florian Weber, Thiemo Wambsganss, Seyed Parsa Neshaei, and Matthias Soellner. 2024. LegalWriter: An intelligent writing support system for structured and persuasive legal case writing for novice law students. In *Proceedings of the CHI conference on human factors in computing systems*. 1–23.
- [35] Kate Williams, Mary Woolliams, and Jane Spiro. 2020. *Reflective writing*. Bloomsbury Publishing.
- [36] Anke Wischgoll. 2017. Improving undergraduates' and postgraduates' academic writing skills with strategy training and feedback. In *Frontiers in education*, Vol. 2. Frontiers Media SA, 33.
- [37] Irmtraud Wolfbauer, Mia Magdalena Bangerl, Katharina Maitz, and Viktoria Pammer-Schindler. 2023. Rebo at work: Reflecting on working, learning, and learning goals with the reflection guidance chatbot for apprentices. In *Extended abstracts of the 2023 CHI conference on human factors in computing systems*. 1–7.
- [38] Daijin Yang, Yanpeng Zhou, Zhiyuan Zhang, Toby Jia-Jun Li, and Ray LC. 2022. AI as an Active Writer: Interaction strategies with generated text in human-AI collaborative fiction writing. In *Joint proceedings of the ACM IUI workshops*, Vol. 10. CEUR-WS Team, 1–11.
- [39] Tan Yu, Anbang Xu, and Rama Akkiraju. 2024. In defense of RAG in the era of long-context language models. *arXiv preprint arXiv:2409.01666* (2024).

Appendix

Prompts Used in Memoire

Prompt Used: Critical Questions.

[System]

You are assisting a student in writing a reflection based on their previous reflections. Based on the reflections retrieved and what the student has written so far, formulate three questions for the student to use to reflect, to ensure coherence, and to maintain a thoughtful and reflective tone. Respond in the same language that the user uses. Do not write anything other than the questions. Provide your answer in a numerated list with one \n character between each question.

[User]

Here is the current reflection the student is writing:

{current text}

Here is the key reflection that is relevant to their current reflection:

{prior reflection}

Prompt Used: Autocomplete Suggestions.

[System]

You are assisting a student by generating the next sentence in their writing, drawing on insights and lessons from their past written reflections. Review the relevant past reflection, focus on overall ideas, and provide a continuation that is coherent and thoughtful. Respond in the same language that the user uses. Only output a single sentence that naturally continues the student's current writing.

[User]

Here is the current reflection the student is writing:

{current text}

Here is the key reflection that is relevant to their current reflection:

{prior reflection}

Please provide the next sentence for the student's writing.

Prompt Used: Summary.

[System]

Help a student recall insights from past reflections for their current writing. Respond in the same language that the user uses. Provide a brief summary of the most relevant past reflection, focusing on key ideas, insights, or lessons to inspire writing their current reflection.

[User]

Here is the current reflection the student is writing:

{current text}

Here is the key reflection that is relevant to their current reflection:

{prior reflection}

Please provide a summary of the past reflection to help the student remember its key points.

Results Obtained for Post-survey Constructs

- Technology Acceptance ($M_1 = 4.68$, $SD_1 = 1.48$)
- Perceived Usefulness ($M_2 = 4.87$, $SD_2 = 1.06$)
- Perceived Ease of Use ($M_3 = 4.85$, $SD_3 = 1.06$)
- Perceived Improvement in Writing ($M_4 = 4.26$, $SD_4 = 1.16$)
- Perceived Improvement in Writing in the Long Run ($M_5 = 4.90$, $SD_5 = 1.13$)
- Correctness of the Suggestions ($M_6 = 4.92$, $SD_6 = 1.08$)
- Excitement After Interaction ($M_7 = 4.51$, $SD_7 = 1.05$)